Task 3: Login System with Validation

Aim

To develop a **Login System** using **HTML5, CSS3, JavaScript, and Database connectivity** that validates user inputs, checks login credentials from the database, and displays **dynamic error or success messages**.

Algorithm

1. Start the program.
2. Create a login page with fields for Email and Password.
3. Design the page using CSS3 for a professional appearance.
4. When the user clicks the login button, read the entered Email and Password.
5. Validate the input fields using JavaScript:
    - Check whether fields are empty.
    - Check whether email format is valid.
    - Check whether password meets required conditions.
6. If validation fails, show dynamic error messages on the page.
7. If validation is successful, send the entered credentials to the server.
8. Connect to the database and search for matching user credentials.
9. If credentials exist in the database, allow login and show success message.
10. If credentials do not match, display “Invalid Email or Password”.
11. Show messages dynamically without reloading unnecessarily.
12. Stop the program.

Program:

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Login System</title>
  <link rel="stylesheet" href="style.css" />
</head>
<body>
```

```
  <div class="login-container">
    <div class="login-card">
      <h2>Welcome Back</h2>
      <p class="subtitle">Login to continue</p>
      <form id="loginForm">
        <div class="input-group">
          <label for="email">Email Address</label>
          <input type="email" id="email" placeholder="Enter your email" />
          <small class="error" id="emailError"></small>
        </div>
        <div class="input-group">
          <label for="password">Password</label>
          <input type="password" id="password" placeholder="Enter your password" />
          <small class="error" id="passwordError"></small>
        </div>
        <button type="submit">Login</button>
        <p id="message"></p>
      </form>
    </div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Script.js:

```
const form = document.getElementById("loginForm");
const email = document.getElementById("email");
const password = document.getElementById("password");
const emailError = document.getElementById("emailError");
const passwordError = document.getElementById("passwordError");
const message = document.getElementById("message");
form.addEventListener("submit", async function (e) {
```

```
e.preventDefault();
emailError.textContent = "";
passwordError.textContent = "";
message.textContent = "";
let isValid = true;
if (email.value.trim() === "") {
    emailError.textContent = "Email is required";
    isValid = false;
} else if (!validateEmail(email.value.trim())) {
    emailError.textContent = "Enter a valid email address";
    isValid = false;
}
if (password.value.trim() === "") {
    passwordError.textContent = "Password is required";
    isValid = false;
} else if (password.value.trim().length < 6) {
    passwordError.textContent = "Password must be at least 6 characters";
    isValid = false;
}
if (!isValid) {
    return;
}
try {
    const response = await fetch("/login", {
        method: "POST",
        headers: {
            "Content-Type": "application/json"
        },
        body: JSON.stringify({
            email: email.value.trim(),
            password: password.value.trim()
```

```
        })
    });
    const data = await response.json();
    if (data.success) {
        message.style.color = "#86efac";
        message.textContent = data.message;
    } else {
        message.style.color = "#fca5a5";
        message.textContent = data.message;
    }
  } catch (error) {
    message.style.color = "#fca5a5";
    message.textContent = "Server connection failed";
  }
});
function validateEmail(email) {
  const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  return pattern.test(email);
}
```

Style.css:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: "Segoe UI", sans-serif;
}
body {
  min-height: 100vh;
  background: linear-gradient(135deg, #0f172a, #1e293b, #334155);
  display: flex;
  justify-content: center;
```

```
  align-items: center;
   padding: 20px;
}
.login-container {
   width: 100%;
   max-width: 420px;
}
.login-card {
   background: rgba(255, 255, 255, 0.08);
   backdrop-filter: blur(16px);
   -webkit-backdrop-filter: blur(16px);
   border: 1px solid rgba(255, 255, 255, 0.15);
   border-radius: 22px;
   padding: 35px 30px;
   box-shadow: 0 12px 35px rgba(0, 0, 0, 0.35);
   color: white;
}
.login-card h2 {
   text-align: center;
   font-size: 30px;
   margin-bottom: 8px;
}
.subtitle {
   text-align: center;
   color: #cbd5e1;
   margin-bottom: 25px;
   font-size: 14px;
}
.input-group {
   margin-bottom: 18px;
}
```

```
.input-group label {
    display: block;
    margin-bottom: 8px;
    font-size: 14px;
    color: #e2e8f0;
}
.input-group input {
    width: 100%;
    padding: 13px 14px;
    border-radius: 12px;
    border: 1px solid rgba(255,255,255,0.15);
    outline: none;
    background: rgba(255,255,255,0.10);
    color: white;
    font-size: 15px;
    transition: 0.3s;
}
.input-group input::placeholder {
    color: #cbd5e1;
}
.input-group input:focus {
    border-color: #38bdf8;
    box-shadow: 0 0 10px rgba(56, 189, 248, 0.35);
}
button {
    width: 100%;
    padding: 13px;
    border: none;
    border-radius: 12px;
    background: linear-gradient(135deg, #38bdf8, #0ea5e9);
    color: white;
```

```
    font-size: 16px;
    font-weight: 700;
    cursor: pointer;
    transition: 0.3s;
    margin-top: 8px;
}
button:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 18px rgba(14, 165, 233, 0.35);
}
.error {
    display: block;
    margin-top: 6px;
    color: #fca5a5;
    font-size: 13px;
    min-height: 16px;
}
#message {
    text-align: center;
    margin-top: 16px;
    font-size: 14px;
    font-weight: 600;
    min-height: 20px;
}
```

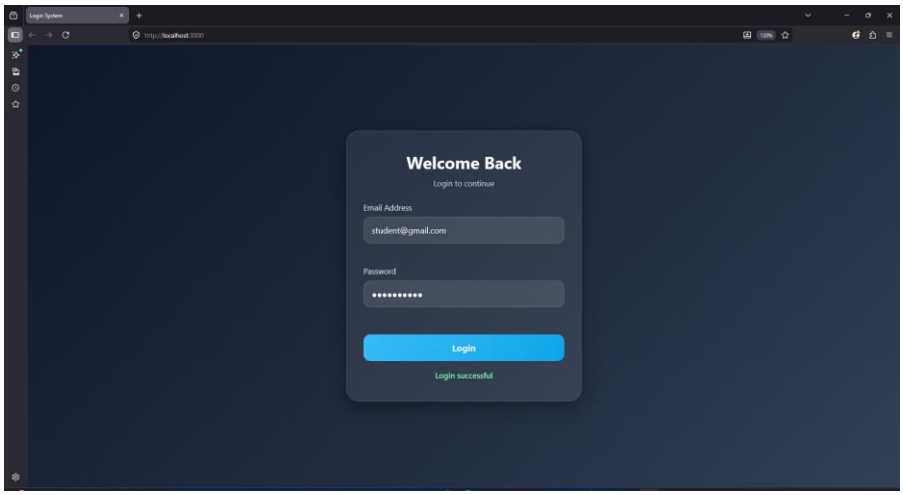
Server.js:

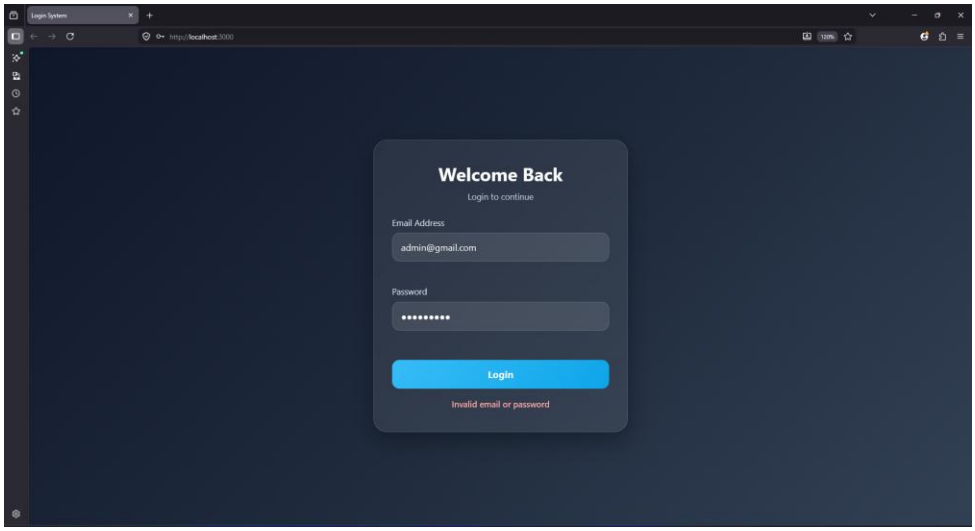
```
const express = require("express");
const mysql = require("mysql2");
const cors = require("cors");
const path = require("path");
const app = express();
```

```
app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, "public")));
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "tehazeeb@2005",
  database: "login_db"
});
db.connect((err) => {
  if (err) {
    console.log("Database connection failed:", err.message);
  } else {
    console.log("Connected to MySQL database");
  }
});
app.post("/login", (req, res) => {
  const { email, password } = req.body;
  if (!email || !password) {
    return res.json({
      success: false,
      message: "Email and password are required"
    });
  }
  const sql = "SELECT * FROM users WHERE email = ? AND password = ?";
  db.query(sql, [email, password], (err, result) => {
    if (err) {
      return res.json({
        success: false,
        message: "Database error"
      });
```

```
        }
        if (result.length > 0) {
            return res.json({
                success: true,
                message: "Login successful"
            });
        } else {
            return res.json({
                success: false,
                message: "Invalid email or password"
            });
        }
    });
});
const PORT = 3000;
app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```

Outputs:





Result:

To develop a **Login System** using **HTML5, CSS3, JavaScript, and Database connectivity** that validates user inputs, checks login credentials from the database, and displays **dynamic error or success messages** is executed successfully.